Table 12. Hierarchical Pruning with S_d=4 on Cora

Methods	F-CK	F-BD	F-KW	F-GD	MAJORITY-F
Precision (%)	**91.67**	86.30	86.30	84.93	87.14
Recall (%)	64.71	61.76	61.76	60.78	59.80
Cost Reduction	60%	60%	60%	60%	60%

the entire ensemble for Cora). Interestingly, the F-CK pruning for Cora achieves the best ensemble pruning performance in terms of the highest precision of 91.67% and highest recall of 64.71%, indicating the diverse utilities of different focal diversity pruning methods for different datasets. Therefore, the focal diversity consensus voting based ensemble pruning is beneficial for delivering more stable and consistent ensemble pruning efficiency in terms of precision and robustness.

Table 13. Hierarchical Pruning with S_d=3 on MNIST

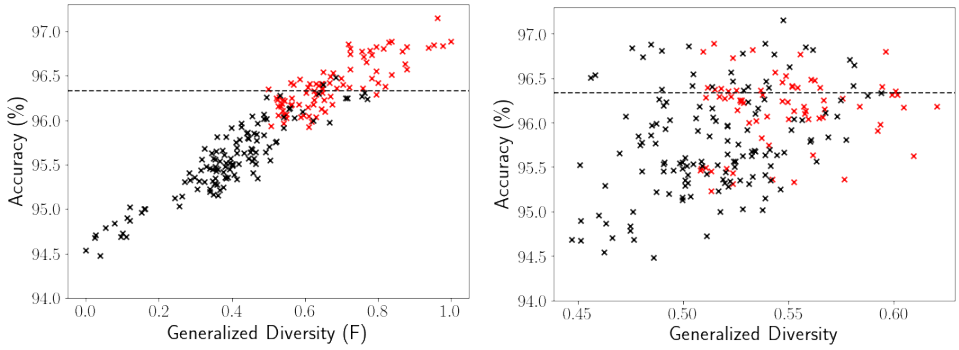
Methods	F-CK	F-BD	F-KW	F-GD	MAJORITY-F
Precision (%)	**100**	75	75	75	75
Recall (%)	60	60	60	60	60
Cost Reduction	57%	57%	57%	57%	57%

MNIST. We then use popular machine learning models, such as KNN, SVM and Logistic Regression in Table 4 to evaluate the effectiveness of our focal diversity based hierarchical pruning methods on MNIST. Table 13 shows the results with $\beta = 50\%$ and $S_d = 3$. We highlight two interesting observations. *First,* all focal diversity based pruning methods achieved over 75% precision in identifying sub-ensembles with the same or improved ensemble accuracy over 96.36%, which is the reference accuracy of the entire ensemble for MNIST. It demonstrates that our focal diversity based hierarchical pruning methods are also effective on representative machine learning models. *Second,* among the Top-3 selected ensemble teams by F-GD or F-CK, the ensemble team 346 (RBF SVM, Random Forest and Neural Network), achieved the highest accuracy of 97.04%, significantly outperforming the entire ensemble of 7 machine learning models with 96.36% accuracy and reducing the ensemble execution cost by over 57%. Moreover, given the Random Forest is already an ensemble of decision trees, it shows that integrating ensemble models, such as the Random Forest, in an ensemble team can further improve the overall predictive performance.

5.4 Focal vs. Baseline Diversity in Hierarchical Pruning

We next compare our focal diversity metrics with baseline diversity metrics by using the same hierarchical pruning algorithm. Figure 5 shows the comparison of our focal F-GD diversity (Figure 5a) and the baseline GD diversity (Figure 5b) in hierarchical pruning for the ensemble teams of size $S = 4$. The black dots mark the ensembles that are pruned out by the hierarchical pruning while the red ones represent the remaining selected ensembles. The reference ensemble accuracy 96.33% of the entire deep ensemble on CIFAR-10 is marked by the horizontal black dashed line. Overall, the hierarchical pruning with our focal F-GD diversity metric achieved much better performance than the baseline GD diversity metric. There are two primary reasons behind this observation. *First,* our focal diversity metrics can better capture the failure independence of member networks of a deep ensemble than baseline diversity metrics. Therefore, when pruning out a low focal diversity branch, such as in Figure 3d with $S = 3$, most ensembles of a larger size with low diversity (with low F-GD scores) will also be pruned out in Figure 3e (same as Figure 5a) with $S = 4$. *Second,* focal diversity

metrics are more effectively correlated to ensemble accuracy than baseline diversity metrics. Therefore, by pruning out low diversity ensembles, our focal diversity metrics can successfully identify high performance ensemble teams with high ensemble accuracy and low ensemble execution cost.



(a) Focal Diversity (F-GD) (b) Baseline Diversity (GD)

Fig. 5. Hierarchical Pruning with F-GD and GD for Ensemble Teams of $S = 4$ (CIFAR-10, $\beta = 10\%, S_d = 5$)

Table 14. Comparing Top-3 Ensemble Teams of S=4 by F-GD and GD (CIFAR-10, $\beta = 10\%, S_d = 5$) with the 96.33% accuracy of the entire ensemble of 10 models shown in Table 3

Method	F-GD			GD		
Ensemble Team	0235	0234	0123	0268	0456	1345
Ensemble Acc (%)	96.89	96.84	97.15	96.18	95.63	96.17
Acc Improv (%) (Over 96.33%)	**0.56**	**0.51**	**0.82**	-0.15	-0.70	-0.16

Table 15. Comparing Top-3 Ensemble Teams of S=5 by F-GD and GD (ImageNet, $\beta = 10\%, S_d = 5$) with the 96.33% accuracy of the entire ensemble of 10 models shown in Table 3

Method	F-GD			GD		
Ensemble Team	12345	23459	23458	03478	02356	01257
Ensemble Acc (%)	80.77	80.50	80.44	78.53	79.25	79.19
Acc Improv (%) (Over 79.82%)	**0.95**	**0.68**	**0.62**	-1.29	-0.57	-0.63

Table 14 lists the Top-3 sub-ensembles selected by F-GD and GD in Figure 5. All the Top-3 sub-ensembles selected by our hierarchical pruning using focal diversity F-GD improve the reference accuracy of 96.33% achieved by the entire ensemble on CIFAR-10. In comparison, the Top-3 sub-ensembles chosen by the baseline GD diversity all result in ensemble accuracy lower than 96.33%, the reference accuracy of the entire ensemble on CIFAR-10, failing to meet the ensemble pruning objective. Similar observations can be found for the ImageNet dataset in Table 15.

The empirical results on all four benchmark datasets demonstrate that our focal diversity based hierarchical pruning framework and algorithms are effective in identifying and selecting space

and time efficient sub-ensembles from a given large ensemble team, while offering competitive ensemble accuracy.

Table 16. Execution Time (s) Comparison of Baseline and Hierarchical Ensemble Pruning

Execution Time (s)		CIFAR-10	ImageNet
Baseline (No Pruning)	BD	0.90	79.08
	GD	0.84	75.83
	F-BD	2.01	97.26
	F-GD	2.01	93.51
Hierarchical Pruning	F-BD	**0.63**	**21.89**
	F-GD	**0.62**	**21.79**

5.5 Execution Time Comparison

We show the execution time of our hierarchical ensemble pruning based on F-BD and F-GD metrics in Table 16 for CIFAR-10 and ImageNet. The baseline is no pruning, where a diversity score, such as GD and F-GD, is computed for each possible ensemble team. We highlight three interesting observations. *First,* compared to the baseline focal diversity metrics, F-BD and F-GD, our hierarchical pruning significantly accelerates the process of identifying high-quality sub-ensembles by 3.2∼4.4×, substantially improving the ensemble pruning efficiency. Table 17 presents the execution time breakdown for each team size S from 2 to 5 using the baseline no pruning and hierarchical ensemble pruning. We list the number of all sub-ensemble teams for each team size in the 2nd column (#All Teams). For the baseline no pruning, we show the F-GD diversity computation time for all sub-ensemble teams in the 3rd column for CIFAR-10 and 8th column for ImageNet. For the hierarchical pruning approach, we list the execution time for calculating F-GD scores (4th column for CIFAR-10 and 9th column for ImageNet), the number of teams that participate in the diversity computation (5th column for CIFAR-10 and 10th column for ImageNet), and the reduction in both execution time and #Teams (6th & 7th columns for CIFAR-10 and 11th & 12th columns for ImageNet). Our hierarchical pruning approach will examine all the 45 smallest ensembles without #Teams reduction for S=2. The slight execution time reduction (0.05s, 3.65%) on ImageNet for S=2 can be attributed to the random measurement errors. For both CIFAR-10 and ImageNet, the execution time reduction and #Teams reduction by our hierarchical pruning closely match with each other, such as 89.95% time reduction and 89.29% of the teams pruned out for S=5 on ImageNet. This indicates that the overall performance improvement (3.2∼4.4×) of our hierarchical pruning originates from the pruned teams that will not participate in diversity computation and thus reduce the execution time. *Second,* for the baseline without pruning, our focal diversity metrics, F-BD and F-GD, take longer execution time than BD and GD, which is due to the computation of focal diversity scores involving two steps for computing the focal negative correlation scores and aggregating the S focal negative correlation scores for each sub-ensemble of size S. *Third,* compared to the baseline diversity metrics, BD and GD, our F-BD and F-GD powered hierarchical pruning can reduce the execution time by 26%∼72%, further demonstrating the efficiency of our hierarchical pruning approach.

Compared with the entire ensemble of all 10 base models, our focal diversity based hierarchical pruning approach can effectively reduce ensemble execution costs in terms of #Parameters (storage cost), GFLOPs (computing cost), and inference latency. Table 18 presents the ensemble execution costs for the entire ensemble and 3 Top-1 ensembles for $S = 3, 4, 5$ identified by our F-GD powered hierarchical pruning approach. These three sub-ensembles all achieve higher ensemble accuracy than the large 10-model ensemble with much smaller team sizes, significantly improving the